

NIKHITHA K P

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Objective

Computer Science (AI & ML) undergraduate with hands-on experience in autonomous vehicle perception systems, Flask-based web applications, and backend development using Python. Skilled in ROS2, Autoware, LiDAR point cloud processing, and REST API development. Seeking Software Engineer or AI-focused development roles.

Education

Bachelor of Engineering – Computer Engineering (AI & ML)

Oct 2022 – Jul 2026

KVG College of Engineering (KVGCE), Sullia

CGPA: 8.42

Technical Skills

- **Backend Development:** Python, Flask, Django, REST APIs
- **Frontend Development:** React.js, javascript, HTML5, CSS3, Bootstrap
- **Databases:** MySQL, MongoDB, SQLite
- **AI/ML & Robotics:** ROS2, Autoware Universe, Computer Vision, Gemini API, Pandas, NumPy, Matplotlib
- **Tools & Platforms:** Git, GitHub, Postman, VS Code, Docker, Linux (Ubuntu), WSL2
- **Data Visualization:** Power BI, Tableau, Excel

Experience

Intern – Autonomous Vehicle Systems

Feb 2026 – Present

CSD, NITK Surathkal

- Contributed to autonomous vehicle perception workflows using ROS2 and Autoware Universe in a research and simulation environment.
- Processed LiDAR point cloud data using voxel filtering and clustering techniques for real-time environment perception.
- Evaluated object detection and sensor fusion pipelines using LiDAR and camera data in Autoware.
- Performed simulation, debugging, and testing using Autoware Planning Simulator, Docker, Linux, and WSL2.
- Supported sensor integration and communication synchronization across ROS2-based autonomous vehicle modules.

Projects

Real Estate Management System (SalagaMithra.in)

Dec 2025

- Developed a responsive full-stack real-estate platform using HTML5, CSS3, JavaScript, and Bootstrap.
- Built backend CRUD functionality, user authentication, and admin verification workflows using Flask and Python.
- Integrated MySQL and MongoDB to manage relational user data and flexible property listings efficiently.
- Implemented multi-parameter filtering by district, price, and size to improve property search relevance and database performance.

Disease Diagnosis Chatbot

2024

- Built an AI-powered healthcare chatbot using Python and Flask for symptom analysis and preliminary diagnosis support.
- Integrated the Gemini API with prompt engineering to generate structured and relevant medical responses.
- Designed a responsive front-end interface and implemented backend validation to improve usability and reliability.
- Focused on privacy-aware input handling and clean user interaction flow.

College Placement Tracker

2024

- Developed a web-based placement management system using Flask to streamline data flow between students, recruiters, and admins.
- Designed relational data models in SQLite to support secure authentication and track application status.
- Created an interactive dashboard using HTML5 and CSS3 for real-time placement drive updates and monitoring.

Certifications

- Python for Data Science – IBM SkillsBuild
- Certified Data Analyst – One Roadmap
- SQL for Beginners: Learn SQL using MySQL and Database Design
- Introduction to Generative AI Studio – Simplilearn (Google Cloud)

Languages

English, Kannada, Hindi